

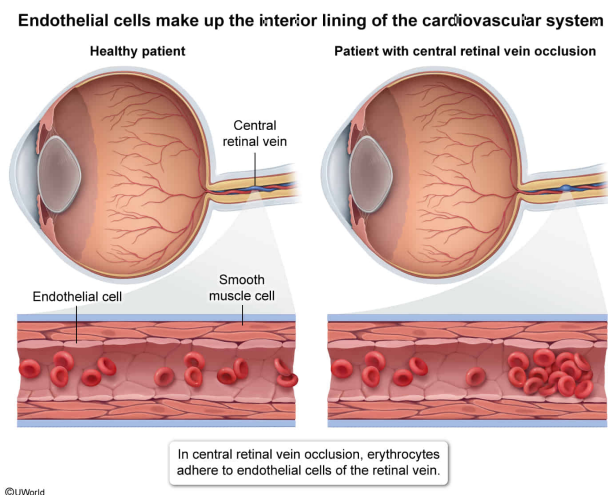
Central retinal vein occlusion (CRVO), a disorder that leads to blindness, is thought to be caused by erythrocytes attaching to the interior lining of the central retinal vein. Erythrocytes most likely adhere to which structures in patients with CRVO?

- ☐ A. Smooth muscle cells [10%]
- ☐ B. Platelets [8%]
- ☐ C. Leukocytes [1%]
- ✓ ☒ D. Endothelial cells [79%]

Correct

78% answered correctly

Explanation:



Endothelial cells are a type of specialized epithelial cell. They form the endothelium, or interior lining of the cardiovascular system (**heart** and blood vessels). The endothelium provides a smooth surface that reduces the friction of **blood** moving through the cardiovascular system. However, substances such as **erythrocytes** may become attached to the endothelium, disrupting the normal flow of blood and causing vascular occlusion (ie, blockage).

In this question, central retinal vein occlusion (CRVO) is described as a disorder that leads to blindness. It is thought to be caused by erythrocytes (red blood cells) attaching to the interior lining of the central retinal vein. The interiors of all cardiovascular components (heart, blood vessels) are lined with endothelial cells. Therefore, in a patient with CRVO, erythrocytes most likely adhere to the **endothelial cells** lining the central retinal vein, causing vein occlusion that leads to blindness.

(Choice A) The endothelium of arteries and veins is surrounded by a layer of **smooth muscle cells**. These cells can contract, causing **vasoconstriction** and

increased blood pressure, or they can relax, causing vasodilation and decreased blood pressure. Because smooth muscle cells *surround* the endothelium, they do *not* make up the interior lining of the central retinal vein.

(Choice B) Platelets are non-nucleated cellular fragments that adhere to injured blood vessel walls, forming a **plug** at the injury site. Platelets generally aggregate and adhere to injury sites but do *not* form the interior lining of veins. Therefore, red blood cells would *not* adhere to platelets in patients with CRVO.

(Choice C) Leukocytes (white blood cells) are cells of the **immune system** that protect the body from infectious agents and foreign antigens. These leukocytes can **adhere** to the endothelial lining and migrate between endothelial cells to exit the bloodstream and travel to affected tissue during infection or injury. However, because leukocytes do *not* make up the interior lining of the central retinal vein, erythrocytes do *not* likely adhere to leukocytes in patients with CRVO.

Educational objective:

Endothelial cells make up the endothelium, or the interior lining of the heart and blood vessels, which are the components of the cardiovascular system.

Time spent:0 sec

Subject:
Biology

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System:
Circulation and Respiration